Weekly Program Learning Summary

Dominic Yeakel | 10/2/2026

Week 1: Networking & Security Foundations

1. **DNS and DHCP:** I learned that DNS translates domain names into IP addresses, while DHCP automatically assigns IP addresses and network settings to devices joining a network. This helped me understand how attackers can abuse these services, through tactics like DNS spoofing or rogue DHCP servers, to redirect or intercept traffic.
2. **Routers and switches:** I learned that routers forward traffic between different networks using IP addresses, while switches connect devices within the same network using MAC addresses. This helped me understand how network layout determines where traffic flows and where security controls like segmentation and monitoring should be placed.
3. **TCP vs UDP:** I learned that TCP provides reliable, ordered delivery of data, while UDP uses less overhead without guaranteeing delivery. This helped me understand how network communication affects the evaluation of security risks.

Week 2: Linux/Windows Administration

1. **Simple Bash commands and flags:** I learned that basic Bash commands like ls, cd, and grep can be adjusted with flags such as -l or -a to show more detail, including hidden files and permissions. This helped me understand how analysts use the command line to quickly inspect a system and spot suspicious files during an investigation.
2. **Linux file system:** I learned that the Linux file system is one hierarchy starting at the root (/), with key directories like /etc for configuration, /var/log for logs, and /home for user data. This helped me understand where to look for evidence and which locations attackers are most likely to target or tamper with.
3. **Windows and Linux hardening:** I learned that hardening Windows and Linux systems means reducing the attack surface by removing unused services, applying updates, enforcing strong account and permission settings, and enabling firewalls. This helped me understand how proactive configuration choices lower security risk before an attack ever happens.

Week 3: Virtualization & Containers

1. **Hypervisor vs supervisor:** I learned that a supervisor is the privileged mode where an operating system's kernel controls a single machine's hardware, while a hypervisor sits beneath one or more guest operating systems and manages how they share physical resources. This helped me understand why compromising a hypervisor is more serious than compromising one OS, since it can expose every virtual machine it hosts.
2. **Type 1 vs Type 2 hypervisors:** I learned that Type 1 hypervisors run directly on the hardware, while Type 2 hypervisors run as an application on top of a host operating

system. This helped me understand that Type 2 setups add the host OS to the attack surface, which is why Type 1 is preferred in enterprise and data center environments.
3. **Common VM and hypervisor threats:** I learned that common threats include VM escape, where an attacker breaks out of a guest into the hypervisor, as well as VM sprawl, insecure snapshots, and poorly isolated virtual networks. This helped me understand why virtual environments need the same patching, access control, and monitoring as physical systems.

Week 4: Network Security & Cryptography

1. **Routing (static vs dynamic) and routing security:** I learned that static routes are configured manually and stay fixed, while dynamic routing protocols like OSPF and BGP let routers learn and share routes automatically. This helped me understand why dynamic routing needs protections such as route authentication and filtering, since attackers can inject false routes to hijack or reroute traffic.
2. **Access control for network device management:** I learned that managing routers and switches securely means using encrypted protocols like SSH instead of Telnet and role-based access with least privilege. This helped me understand that network devices are high-value targets, so controlling who can change their configurations is critical to protecting the whole network.
3. **Symmetric vs asymmetric encryption:** I learned that symmetric encryption uses one shared key for both encryption and decryption, while asymmetric encryption uses a public and private key pair. This helped me understand why secure systems often combine the two, using asymmetric encryption to safely exchange keys and faster symmetric encryption to protect the data itself.

Week 5: Next-Gen Firewalls

1. **Next-gen vs traditional firewalls:** I learned that traditional firewalls filter traffic mainly by IP address, port, and protocol, while next-generation firewalls also inspect the content of traffic and identify the applications and users behind it. This helped me understand why port-based rules alone fall short, since modern threats often hide inside allowed traffic like web browsing.
2. **NGFW capabilities:** I learned that NGFWs combine features such as deep packet inspection, application awareness, intrusion prevention, user identity integration, and threat intelligence in a single device. This helped me understand how layering these controls lets security teams detect and block attacks that a basic firewall would let through.
3. **Default deny:** I learned that a default deny policy blocks all traffic unless a rule explicitly allows it. This helped me understand how following least privilege at the network level shrinks the attack surface and limits what an attacker can reach.

Week 6: Threats & Vulnerabilities

1. **Endpoint detection and prevention:** I learned that endpoint protection tools work in two ways: prevention tools block known threats before they run, while endpoint detection and response (EDR) tools continuously monitor device behavior to spot and contain suspicious activity. This helped me understand why endpoints like laptops and servers need both layers, since attackers often get past preventive controls and must then be caught by their behavior.
2. **Threat intelligence:** I learned that threat intelligence is collected and analyzed information about attackers, their tactics, and indicators of compromise such as malicious IPs, domains, and file hashes. This helped me understand how organizations use this information to anticipate attacks, prioritize defenses, and detect threats faster.
3. **The SOC:** I learned that a Security Operations Center (SOC) is a team that monitors an organization's systems around the clock, using tools like SIEM to triage alerts, investigate incidents, and coordinate response. This helped me understand how people, processes, and technology work together to turn raw security data into timely action against threats.

Week 7: SOC Ops, IR & AI

1. **SOC tiers:** I learned that SOCs are commonly organized into tiers, where Tier 1 analysts triage and validate alerts, Tier 2 analysts investigate deeper and respond to confirmed incidents, and Tier 3 analysts handle advanced threat hunting and complex cases. This helped me understand how escalating alerts by severity lets a SOC handle large alert volumes while saving its most experienced people for the most serious threats.
2. **Playbook vs runbook:** I learned that a playbook outlines the overall strategy, roles, and decision points for responding to a type of incident, while a runbook gives the detailed, step-by-step technical actions to carry out a specific task. This helped me understand how documented procedures make incident response faster, more consistent, and easier to automate.
3. **IR lifecycle:** I learned that the incident response lifecycle, as defined by NIST, moves through preparation; detection and analysis; containment, eradication, and recovery; and post-incident activity. This helped me understand that effective response starts before an attack happens and ends with lessons learned that strengthen future defenses.

Week 8: SOC Tools & Data Analytics

1. **Wireshark:** I learned that Wireshark is a packet analyzer that captures network traffic and lets me filter and inspect individual packets, protocols, and conversations in detail. This helped me understand how analysts use packet-level evidence to confirm suspicious activity, such as malware communication or data exfiltration, that higher-level alerts only hint at.
2. **Splunk, Microsoft Sentinel, and Defender XDR with Security Copilot:** I learned that Splunk and Microsoft Sentinel are SIEM platforms that collect and search log data across an environment, while Defender XDR correlates signals across endpoints, email,

identities, and cloud apps, with Security Copilot adding AI assistance for summarizing and investigating incidents. This helped me understand how these tools give SOC analysts a centralized view of threats and speed up investigations.

3. **How SIEM, EDR, SOAR, and AI fit together:** I learned that SIEM centralizes and correlates log data, EDR monitors and responds on endpoints, SOAR automates response workflows, and AI helps prioritize alerts and accelerate analysis. This helped me understand that these tools work best as one connected system, turning large volumes of security data into faster, more consistent detection and response

Week 9: Power BI for Security and Fraud Analytics

1. **Power BI Desktop and Service:** I learned that Power BI Desktop is the application used to connect to data, build data models, and design reports, while the Power BI Service is the cloud platform used to publish, share, and refresh those reports. This helped me understand how security and fraud findings move from an analyst's workstation to dashboards that decision-makers can monitor.
2. **Sensitivity labels and row-level security:** I learned that sensitivity labels classify and protect reports based on how confidential their data is, while row-level security limits which rows of data each user can see. This helped me understand how to share security and fraud dashboards widely without exposing sensitive data to people who shouldn't have access.
3. **Slicers vs filters:** I learned that slicers are visible, on-canvas controls that let report users interactively narrow the data, while filters are set in the filter pane and can apply to a single visual, a page, or the entire report. This helped me understand how to design dashboards that let analysts quickly drill into suspicious activity while keeping the underlying report logic consistent